

Lecture 19

The thermodynamic limit: densities, ground states, excitations

Objectives. Until now a Bethe state has been a finite list of roots. Let $N \rightarrow \infty$ at fixed filling and the list becomes a density, the Bethe equations one linear integral equation for it. Solving that equation is this lecture: by Fourier transform it gives the Heisenberg antiferromagnet's ground-state energy $-J \ln 2$ exactly, and numerically it gives the Lieb–Liniger gas its equation of state from weak coupling to the impenetrable limit, where it turns back into free fermions. Removing one root from the sea leaves two holes of spin $\frac{1}{2}$: the spinons, whose continuum neutron scattering sees. Densities, dressing and the dressed velocity are what [Lecture 20](#) runs at finite temperature and [Lecture 23](#) out of equilibrium.

19.1 From roots to densities

[Lecture 15](#) wrote the Bethe equations of the Heisenberg chain in logarithmic form (15.18),

$$N\theta_1(\lambda_j) - \sum_l \theta_2(\lambda_j - \lambda_l) = 2\pi I_j, \quad \theta_n(\lambda) = 2 \arctan \frac{2\lambda}{n}, \quad I_j \in \mathbb{Z} + \frac{N-M+1}{2}, \quad (19.1)$$

where the sum may run over all l because $\theta_2(0) = 0$. It also introduced, in a tutorial, the *counting function*; everything below is built on it, so here it is again:

$$\zeta(\lambda) = \theta_1(\lambda) - \frac{1}{N} \sum_l \theta_2(\lambda - \lambda_l), \quad \frac{N\zeta(\lambda_j)}{2\pi} = I_j. \quad (19.2)$$

Built from the roots and then read as a function of a free λ , it passes the value I_j where the j th root sits ([Figure 19.1](#), left); the values it passes where there is *no* root are the *holes*.

How many values may I_j take? Since $\theta_n(\pm\infty) = \pm\pi$, the function (19.2) runs from $-(N-M)\pi/N$ to $+(N-M)\pi/N$, so $N\zeta/2\pi$ sweeps an interval of length $N-M$.

Proposition 19.1 (The vacancy count). For M finite real roots there are exactly

$$\#\{\text{vacancies}\} = N - M \quad (19.3)$$

allowed values of I : the $N-M$ elements of $\mathbb{Z} + \frac{N-M+1}{2}$ with $|I| \leq \frac{N-M-1}{2}$. If moreover the roots are pairwise distinct and ζ is strictly increasing, they occupy M distinct ones and $N-2M$ are left as holes.

Proof. The allowed values are those strictly inside the range of $N\zeta/2\pi$, an interval of length $N-M$ centred at 0; the lattice $\mathbb{Z} + \frac{N-M+1}{2}$ meets its interior in $N-M$ points, symmetric about 0 ([Lecture 15](#) checked the endpoints: $I = \pm \frac{N-M+1}{2}$ belongs to a root at infinity). A strictly

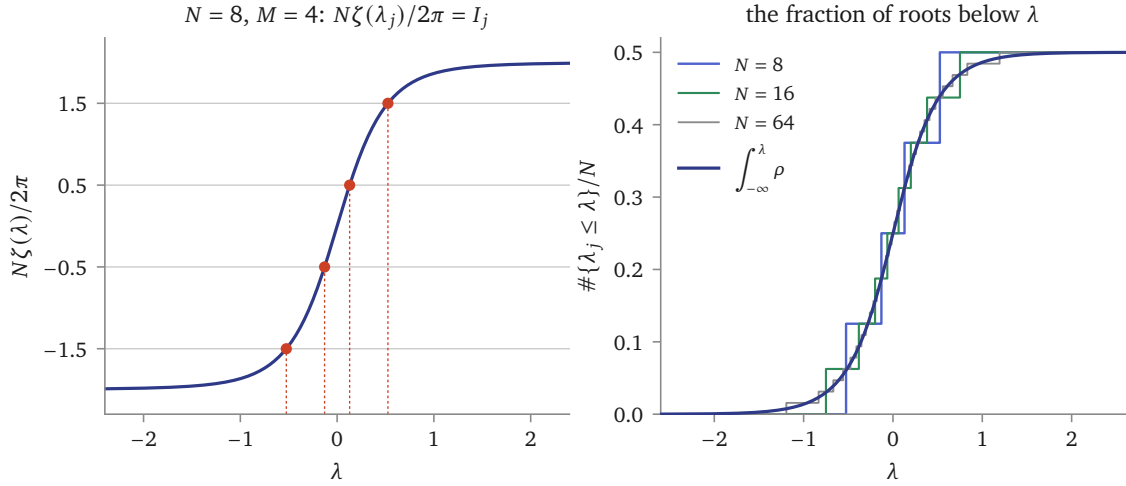


Figure 19.1: Left: the counting function of the $N = 8, M = 4$ ground state. Its crossings of the levels I_j are the roots. Right: the inverse picture — the fraction of roots below λ , a staircase of step $1/N$, at $N = 8, 16, 64$. As N grows the staircase becomes the smooth $\int_{-\infty}^{\lambda} \rho$, and ρ is the object this lecture computes.

increasing ζ takes each value at most once, so pairwise distinct roots carry M distinct values of I_j ; subtracting them leaves $N - M - M$. ■

Remark 19.2 (Monotonicity is a hypothesis, not a theorem). Distinct roots need *not* carry distinct quantum numbers: at $N = 22$ two real roots of a genuine eigenstate share $I = \frac{19}{2}$, with $\zeta' < 0$ between them (the tutorial). So the second sentence of [Proposition 19.1](#) is a hypothesis about the branch one is on — the branch this lecture uses, where $\zeta' = 2\pi\rho_t > 0$ by construction. [Lecture 15](#) was careful for the same reason.

Three consequences are worth naming at once, because the whole of [§ 19.5](#) rests on them. On that branch, $M = N/2$ leaves no holes at all: the roots fill every vacancy. $M = N/2 - 1$ leaves two. And $N - 2M$ is even for every M when N is even, so on an even ring holes are created in pairs; a single hole needs odd N .

The limit

Now let $N \rightarrow \infty$ with M/N fixed and suppose the roots and the holes fill out densities: the number of roots in $[\lambda, \lambda + d\lambda]$ is $N\rho(\lambda)d\lambda$, the number of holes $N\rho_h(\lambda)d\lambda$. Then $\frac{1}{N} \sum_l \theta_2(\lambda - \lambda_l) \rightarrow \int \theta_2(\lambda - \mu)\rho(\mu)d\mu$, and $N\zeta/2\pi$ increases by one across every vacancy, so $\zeta'/2\pi$ is the density of vacancies — roots and holes together. Differentiating [\(19.2\)](#) gives the lecture's equation.

Theorem 19.3 (The root-density equation). Write $a_n(\lambda) = \theta'_n(\lambda)/2\pi = \frac{1}{\pi} \frac{n/2}{\lambda^2 + n^2/4}$. Under the assumption above,

$$\rho(\lambda) + \rho_h(\lambda) = a_1(\lambda) - \int_{\mathbb{R}} a_2(\lambda - \mu) \rho(\mu) d\mu. \quad (19.4)$$

• proved from the assumption; for the assumption as the sources make it see [Takahashi §§3.3–3.4](#) and [Korepin–Bogoliubov–Izergin §§1.2–1.3](#)

The assumption is an interchange of limits, and it is not proved here: *say so aloud*. What is assumed is that the chosen solutions of (19.1) converge, that their empirical measure has a density, and that the convergence is good enough to pass inside the integral. It is the subject's standard working hypothesis, tested numerically rather than proved; Figure 19.2 is one such test.

One equation, two models — and a sign

Lecture 16 wrote the Lieb–Liniger equations in the same shape (16.11), $k_j L + \sum_l \theta(k_j - k_l) = 2\pi I_j$ with $\theta(k) = 2 \arctan(k/c)$. Both models are therefore instances of

$$L p_0(\lambda_j) + \sum_l \varphi(\lambda_j - \lambda_l) = 2\pi I_j, \quad \rho + \rho_h = \frac{p'_0}{2\pi} + \frac{1}{2\pi} \int K(\lambda - \mu) \rho(\mu) d\mu, \quad K = \varphi', \quad (19.5)$$

with L the number of sites N or the length of the box. The dictionary between them is

$$\begin{aligned} \text{XXX: } p_0 &= \theta_1, \quad \varphi = -\theta_2, \quad K(\lambda) = \frac{-2}{1 + \lambda^2}, \\ \text{LL: } p_0 &= k, \quad \varphi = +\theta, \quad K(k) = \frac{2c}{k^2 + c^2}. \end{aligned} \quad (19.6)$$

Remark 19.4 (The kernel's sign). The two kernels have opposite signs, and every formula below inherits it. The reason is bookkeeping, not physics: Lecture 15 measures a magnon by $\theta_1 = \pi - k$, which runs *backwards* against the momentum, while Lecture 16 measures a particle by k . In k the chain's equation is $Nk_j + \sum \theta_2 = 2\pi n_j + (M - 1)\pi$ (15.20), with the gas's sign. Flipping the chain's sign is not harmless: § 19.2's Fourier solution would become $\hat{\rho} = e^{-|\omega|/2}/(1 - e^{-|\omega|})$, which diverges at $\omega = 0$ — no normalisable density at all.

19.2 The Heisenberg antiferromagnet

Take $J > 0$ in $H = J \sum (\mathbf{S}_n \cdot \mathbf{S}_{n+1} - \frac{1}{4})$ and N even. A Bethe state has $E = -\sum_j 2J/(4\lambda_j^2 + 1)$ — (15.16), and equally what $\ln \Lambda|_0$ gave in Lecture 18 (Proposition 18.7). Every term is *negative*, which suggests, without proving it (adding a root moves all the others), that the ground state carries as many finite real roots as it can. The largest M is $N/2$ (Lecture 15: the Bethe states are highest weight), and by Proposition 19.1 that sector has exactly $N/2$ vacancies. *The antiferromagnetic ground state fills all of them*: it is the state with $M = N/2$ and no holes, a total singlet — *cited for the identification with the lowest level: Hulthén (1938); Takahashi §3.3; confirmed here against exact diagonalisation, $N \leq 16$* . For odd N the lowest level is made of spin- $\frac{1}{2}$ multiplets — two of them for $N = 3, \dots, 9$ (Lecture 15), so that already at $N = 3$ the lowest level is four-fold degenerate — and the thermodynamic limit is therefore taken along even N .

No holes means $\rho_h \equiv 0$ on the whole line, and (19.4) becomes a convolution equation on $\mathbb{R}$:

$$\rho(\lambda) + \int_{\mathbb{R}} a_2(\lambda - \mu) \rho(\mu) d\mu = a_1(\lambda). \quad (19.7)$$

A convolution on the line is a product after a Fourier transform, and the transform of a_n is as simple as can be.

Theorem 19.5 (The ground-state density and energy). With $\hat{f}(\omega) = \int f(\lambda) e^{i\omega\lambda} d\lambda$ one has

$\hat{a}_n(\omega) = e^{-n|\omega|/2}$, and (19.7) has exactly one solution in $L^1(\mathbb{R})$,

$$\hat{\rho}(\omega) = \frac{1}{2 \cosh(\omega/2)}, \quad \rho(\lambda) = \frac{1}{2 \cosh \pi \lambda}. \quad (19.8)$$

Its energy per site is

$$e_0 = \frac{E}{N} = - \int_{\mathbb{R}} \frac{2J}{4\lambda^2 + 1} \rho(\lambda) d\lambda = -J \ln 2. \quad (19.9)$$

Proof. The function a_n is a Cauchy density of half-width $n/2$, whose transform is $e^{-n|\omega|/2}$ (a residue at $\lambda = \pm in/2$, according to the sign of ω). Transforming (19.7), $\hat{\rho}(1 + e^{-|\omega|}) = e^{-|\omega|/2}$, so $\hat{\rho} = 1/(e^{|\omega|/2} + e^{-|\omega|/2})$, which is (19.8); inverting, $\frac{1}{4\pi} \int_{\mathbb{R}} \frac{e^{-i\omega\lambda}}{\cosh(\omega/2)} d\omega = \frac{1}{2 \cosh \pi \lambda}$. If two L^1 solutions differed, the difference would have a continuous transform killed by $1 + e^{-|\omega|} > 0$, hence zero. For the energy, note that $2J/(4\lambda^2 + 1) = \pi J a_1(\lambda)$, so by Parseval

$$e_0 = -\pi J \int a_1 \rho d\lambda = -\frac{\pi J}{2\pi} \int_{\mathbb{R}} \frac{e^{-|\omega|/2}}{2 \cosh(\omega/2)} d\omega = -\frac{J}{2} \int_0^\infty \frac{2e^{-\omega}}{1 + e^{-\omega}} d\omega = -J \ln 2. \quad \blacksquare$$

Two checks: $\int \rho = \hat{\rho}(0) = \frac{1}{2} = M/N$, so the density knows the filling; and a finite chain's roots really do distribute themselves like (19.8) (Figure 19.2), its energies converging to $-J \ln 2$ along a straight line in $1/N^2$ (Figure 19.3) whose slope the reading explains.

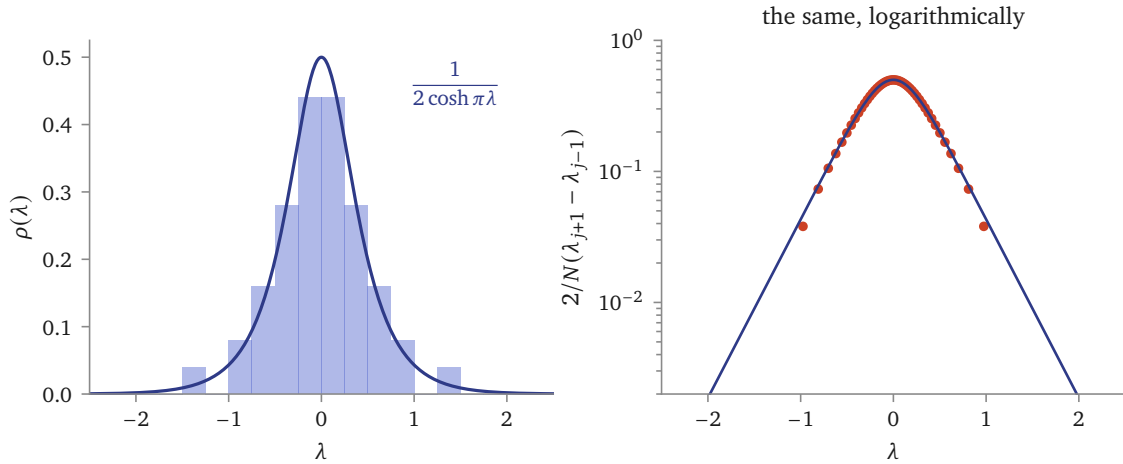


Figure 19.2: $N = 100$, $M = 50$: the 50 Bethe roots of the ground state, solved from (19.1) by Newton's method, as a histogram (left) and through their local spacing $2/N(\lambda_{j+1} - \lambda_{j-1})$ on a logarithmic axis (right), against $1/(2 \cosh \pi \lambda)$. The interior agreement is better than 2% at $N = 100$.

The sea of (19.8) occupies the whole rapidity line, and that is not a paradox: $k = \pi - \theta_1(\lambda)$ carries $\mathbb{R}$ onto the compact $(0, 2\pi)$, so an infinite sea of rapidities is a bounded set of momenta. Rapidity is the variable in which the equations are linear; momentum is the variable the experiment measures. The massless XXZ chain, $|\Delta| < 1$, has the same structure with π replaced by an η -dependent scale · cited: Takahashi ch. 4; Franchini §4.4.

19.3 The Lieb–Liniger gas

For the gas of Definition 16.2, $H = -\sum \partial_j^2 + 2c \sum_{i < j} \delta$ with $c > 0$ on a ring of length L , one thing changes: the energy $E = \sum k_j^2$ is a sum of *positive* terms, so the ground state should

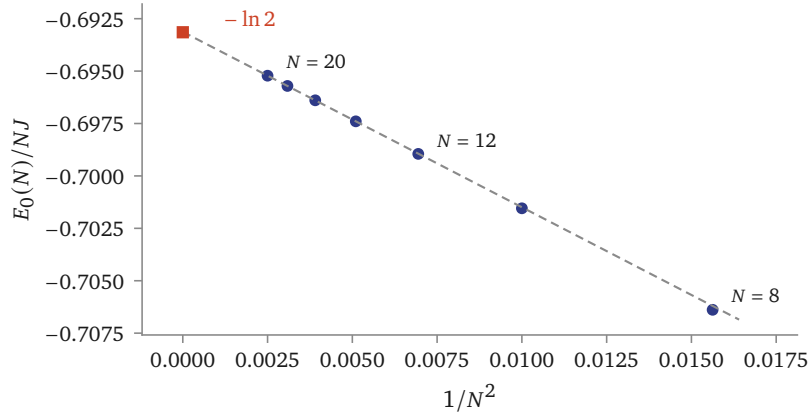


Figure 19.3: Ground-state energy per site from exact diagonalisation, $N = 8, 10, \dots, 20$, against $1/N^2$. The straight line through the largest four extrapolates to $-\ln 2$ within 2×10^{-5} (square); its slope is -0.8368 , within 2% of the asymptotic $-\pi^2/12 = -0.8225$ that the reading on finite size explains. The gap is the higher finite-size corrections, still visible at $N = 20$.

be the one whose quantum numbers sit as close to zero as they can: the consecutive block $I_j = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$ of [Lecture 16](#), which is the ground state at every $c > 0$ · cited: [Yang–Yang \(1969\)](#). Their roots are real and ordered ([Proposition 16.4](#), [Proposition 16.5](#)) and fill an interval $[-q, q]$: a *Fermi sea* of rapidities, with no holes inside and no roots outside. So (19.5) becomes an equation on a finite interval.

Theorem 19.6 (Lieb’s equation). *In the ground state the rapidity density satisfies*

$$\rho(k) = \frac{1}{2\pi} + \frac{1}{2\pi} \int_{-q}^q K(k - k') \rho(k') dk', \quad K(k) = \frac{2c}{k^2 + c^2} = -i \frac{d \ln S}{dk}, \quad (19.10)$$

with q fixed by the density, and then

$$n = \int_{-q}^q \rho dk, \quad \frac{E}{L} = \int_{-q}^q k^2 \rho dk, \quad \gamma = \frac{c}{n}, \quad \frac{E}{N} = n^2 e(\gamma). \quad (19.11)$$

The dimensionless e depends on c and q only through γ .

Proof. All but the last sentence is (19.5) with $p'_0 = 1$, $\rho_h = 0$ on $[-q, q]$ and $\theta' = K = -i d \ln S / dk$ ([Proposition 16.1](#)). For the last, substitute $k = qx$, $g(x) = \rho(qx)$: with $\alpha = c/q$, $g(x) = \frac{1}{2\pi} + \frac{1}{2\pi} \int_{-1}^1 \frac{2\alpha}{(x-y)^2 + \alpha^2} g(y) dy$, which involves α alone. Then $n = q \int g$ and $\gamma = c/n = \alpha / \int g$, and $E/Ln^3 = \int x^2 g / (\int g)^3$: both are functions of α alone, so the pair (γ, e) traces one curve as α runs over $(0, \infty)$, and that curve is a graph because γ is strictly monotone: at fixed c the Neumann series for (19.10) has a positive source and a positive kernel and converges, so enlarging $[-q, q]$ raises ρ pointwise and adds interval; $n(q)$ is strictly increasing, $\gamma = c/n$ strictly decreasing in q , hence strictly increasing in $\alpha = c/q$. ■

Equation (19.10) has no closed-form solution, and this is where the subject turns numerical: a quadrature rule on $[-q, q]$ turns a Fredholm equation of the second kind into an $n \times n$ linear system. How large n must be is set by the one parameter of the proof above. The kernel has width c and the interval width q , so the rule resolves it easily while $\alpha = c/q$ is of order one — twenty points give machine precision at $\gamma \approx 1$ — and must be refined as

$\alpha \rightarrow 0$, where the kernel becomes a spike. At $\gamma \approx 10^{-4}$ a two-hundred-point rule returns a negative density, which is worth knowing before trusting a plot of the weak-coupling end.

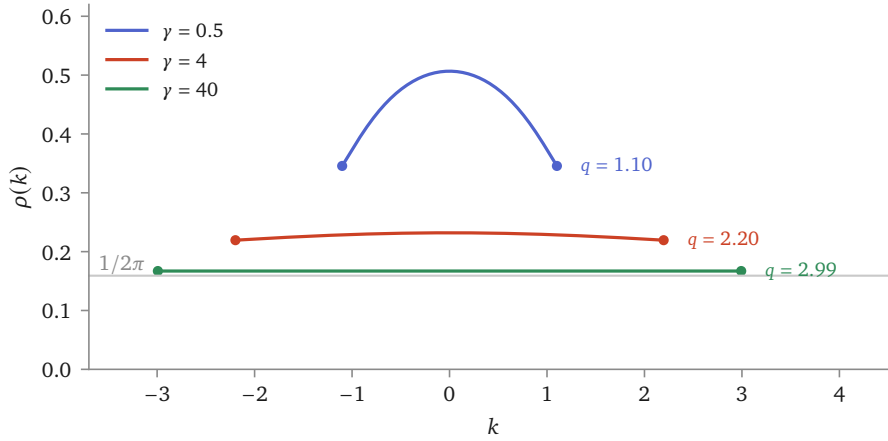


Figure 19.4: The Fermi sea of rapidities at three couplings, all at density $n = 1$. At weak coupling the sea is narrow and ρ is peaked; as γ grows the sea widens towards $q = \pi n$ and ρ flattens onto the free-fermion value $1/2\pi$.

The two limits

Proposition 19.7 (Impenetrable bosons are free fermions). As $\gamma \rightarrow \infty$, $\rho \rightarrow 1/2\pi$ on $[-q, q]$ with $q = \pi n$, and

$$e(\gamma) \rightarrow \frac{\pi^2}{3}, \quad \text{i.e.} \quad \frac{E}{N} \rightarrow \frac{\pi^2 n^2}{3}. \quad (19.12)$$

This is exactly the ground-state energy of N free spinless fermions with $\varepsilon(k) = k^2$ on the same ring.

Proof. In the scaled equation of Theorem 19.6 the kernel integrates to at most $2/\pi\alpha$ over $y \in [-1, 1]$, and $\alpha = c/q \rightarrow \infty$, so $g \rightarrow 1/2\pi$ uniformly. Then $n \rightarrow q/\pi$ and $E/L \rightarrow q^3/3\pi$, whence $e = (E/L)/n^3 \rightarrow \pi^2/3$. For the free fermions the momenta are the N allowed values nearest 0, spaced $2\pi/L$ (Lecture 14), filling $[-\pi n, \pi n]$ at density $1/2\pi$: the same integral. ■

This closes a loop: Lecture 14 solved a model by turning it into free fermions, and here an interacting Bose gas, pushed to infinite repulsion, becomes free fermions — exactly, for every state and not only the ground state, by Girardeau’s map $\psi_B = \prod_{i < j} \text{sgn}(x_i - x_j) \psi_F$ · cited: Girardeau (1960). On a ring the map carries a boundary condition: taking one coordinate once round reverses its order with the other $N - 1$, so periodic bosons need fermions periodic for odd N and antiperiodic for even N . These are continuum fermions, $\varepsilon = k^2$; Lecture 14’s live on a lattice with its own dispersion. What the two share is the structure, not the Hamiltonian.

At the other end $q \rightarrow 0$, the sea shrinks and the expansion is harder; Bogoliubov’s answer is $e = \gamma - \frac{4}{3\pi} \gamma^{3/2} + O(\gamma^2)$ · sketched: mean field, then the zero-point energy of the Bogoliubov modes; cited for the coefficients, against $e = \frac{\pi^2}{3} (1 - 4/\gamma + O(\gamma^{-2}))$ at the other. Figure 19.5 shows both, and shows that neither survives $\gamma \sim 1$; the reading carries the next orders.

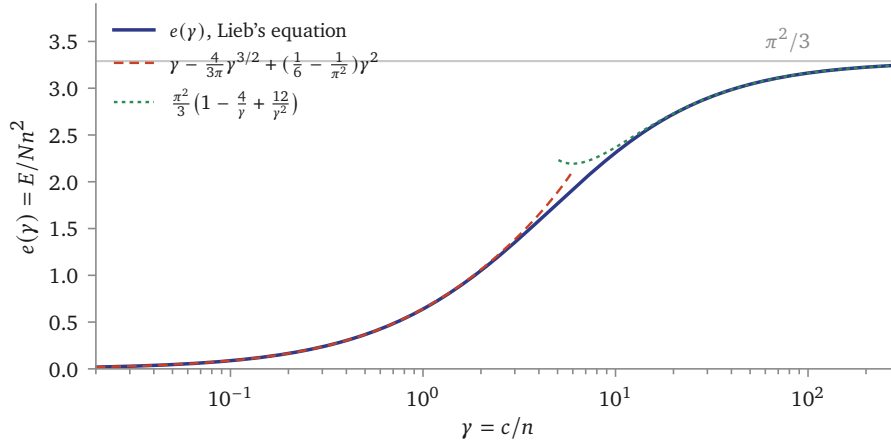


Figure 19.5: The equation of state $e(\gamma)$ from (19.10), with the weak- and strong-coupling series drawn past the range in which they are accurate so that they can be seen leaving the curve. There is no crossover scale: $\gamma \sim 1$ is the whole of it.

19.4 Dressing

Every question asked of a Bethe state in the thermodynamic limit has the same shape. Adding one particle to the sea moves *every other* root a little, so the change in any extensive quantity is the particle's own contribution *plus* the backflow. One linear operator does the bookkeeping for all of them.

Definition 19.8 (Dressing). Let $\vartheta(\lambda) = \rho(\lambda)/\rho_t(\lambda)$ be the filling fraction, $\rho_t = \rho + \rho_h$ (at $T = 0$, the indicator of the sea). The dressed f is the solution of

$$f^{\text{dr}}(\lambda) = f(\lambda) + \frac{1}{2\pi} \int K(\lambda - \mu) \vartheta(\mu) f^{\text{dr}}(\mu) d\mu, \quad (19.13)$$

with K the model's kernel (19.6).

The definition is worth its own name because (19.5) is already an instance of it — and **it is a response, not a total**. Dressing a bare momentum does not give an excitation's momentum: at $c = q = 1$ the dressed $f(k) = k$ has slope 1.2099 at the origin against the excitation's 1.9190. It is the *derivative* that dresses.

Proposition 19.9. $2\pi\rho_t = (p'_0)^{\text{dr}}$ for any ϑ . Now take $T = 0$, so that ϑ is the indicator of the sea, and let an excitation have bare energy ε_0 and bare momentum k_0 ; write $\varepsilon = \varepsilon_0^{\text{dr}}$. If ε vanishes at the edges of the sea, then $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$, and the excitation's dressed velocity is

$$v^{\text{eff}}(\lambda) = \frac{(\varepsilon'_0)^{\text{dr}}(\lambda)}{(k'_0)^{\text{dr}}(\lambda)} = \frac{d\varepsilon}{dp}, \quad p' = (k'_0)^{\text{dr}}. \quad (19.14)$$

Proof (★). Multiply (19.5) by 2π and use $\rho = \vartheta\rho_t$: $2\pi\rho_t = p'_0 + \frac{1}{2\pi} \int K\vartheta(2\pi\rho_t)$, which is (19.13) for $f = p'_0$, for any ϑ . For the second statement, differentiate (19.13) in λ and integrate by parts in μ , using $\partial_\lambda K(\lambda - \mu) = -\partial_\mu K(\lambda - \mu)$:

$$\varepsilon'(\lambda) = \varepsilon'_0(\lambda) + \frac{1}{2\pi} \int K(\lambda - \mu) (\vartheta\varepsilon)'(\mu) d\mu.$$

ϑ being an indicator, $(\vartheta\varepsilon)'$ is $\vartheta\varepsilon'$ *plus* delta functions at the edges weighted by ε there.

Those vanish exactly when ε does, and what is left is (19.13) for ε'_0 . The velocity is then the chain rule. *Without the edge condition the boundary terms survive*, and $\varepsilon' \neq (\varepsilon'_0)^{\text{dr}}$: for the gas at $c = q = 1$, dressing $f(k) = k$ and differentiating gives 1.2099 at $k = 0$ against $(1)^{\text{dr}}(0) = 1.9190$. *The step also needs $T = 0$* : for a smooth ϑ the extra term is $\frac{1}{2\pi} \int K \vartheta' \varepsilon$, a bulk integral that no edge condition kills. ■

Remark 19.10 (Which momentum, and which way it runs). k_0 is the excitation's own bare momentum, not automatically (19.5)'s counting momentum p_0 . For a particle added to the gas the two agree; for the chain they do not, since $p_0 = \theta_1$ runs backwards against $k = \pi - \theta_1$ (Remark 19.4) — and the chain's excitation is not a magnon but a *hole*, of bare energy $-\varepsilon_0$, whose momentum runs *with* ζ . § 19.5 computes it and finds $|\nu^{\text{eff}}| \rightarrow \pi J/2$; reading (19.14) with ε_0 and θ_1 instead flips that sign.

Two things fix the object in place. Equation (19.13) is *linear* with the same kernel for every f , so dressing is one operator $(1 - \frac{1}{2\pi} K \vartheta)^{-1}$: a single numerical solve answers every question. And ϑ carries all the state dependence — sharp at $T = 0$, a smooth Fermi function at finite temperature (Lecture 20), position- and time-dependent out of equilibrium (Lecture 23) — with (19.14) the dressed velocity throughout. What does *not* carry over is the route to it: at $T > 0$ the identity $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ comes from differentiating Yang–Yang's *nonlinear* equation for the pseudo-energy · Lecture 20.

19.5 Spinons, and Lieb's two branches

Return to the antiferromagnet and make the cheapest excitation there is: remove one root. The sector is $M = N/2 - 1$, which by Proposition 19.1 has $N/2 + 1$ vacancies for $N/2 - 1$ roots — two holes, at $\lambda_{h_1}, \lambda_{h_2}$ — while the magnetisation has gone up by one, $\Delta S^z = 1$. Two objects, one unit of S^z between them: by itself that would also allow a spin 1 and a spin 0. What settles it is the multiplet structure — at fixed $\lambda_{h_1}, \lambda_{h_2}$ the states form one triplet and one singlet, exactly as two spin- $\frac{1}{2}$ objects do · cited: *Faddeev–Takhtajan (1981)*. They are the *spinons*, and that is the lecture's most surprising sentence: the elementary excitation of a spin chain is not a magnon of spin 1 but half of one, and it cannot be made alone.

Theorem 19.11 (The spinon). *Let N be even and let a state of the antiferromagnet lie on the branch of Proposition 19.1 — pairwise distinct real roots, ζ increasing — with $M = (N - r)/2$ of them, so that it carries r holes, necessarily an even number, and let r stay fixed as $N \rightarrow \infty$. In the limit of Theorem 19.3, if the holes sit at $\lambda_{h_1}, \dots, \lambda_{h_r}$ then*

$$\Delta E = \sum_i \varepsilon(\lambda_{h_i}), \quad \Delta K = \sum_i p(\lambda_{h_i}) \pmod{2\pi}, \quad (19.15)$$

with

$$\varepsilon(\lambda) = \frac{\pi J}{2 \cosh \pi \lambda}, \quad p(\lambda) = 2\pi \int_{-\infty}^{\lambda} \rho = \frac{\pi}{2} + \arctan \sinh \pi \lambda \in (0, \pi), \quad \varepsilon = \frac{\pi J}{2} \sin p. \quad (19.16)$$

· proved from the first-order form of the assumption noted after the proof; the dispersion is that of des Cloizeaux and Pearson (1962)

Proof (★). With ε_0 as above, $E = N \int \varepsilon_0 \rho$. With holes, (19.4) reads $(1 + a_2^*)\rho = a_1 - \rho_h$, while $(1 + a_2^*)\rho_{(0)} = a_1$ for the sea (19.8); subtracting and putting $\rho_h = \frac{1}{N} \sum_i \delta_{\lambda_{h_i}}$, $\rho - \rho_{(0)} =$

$-\frac{1}{N}(1+a_2^*)^{-1} \sum_i \delta_{\lambda_{h_i}}$. Hence, $1+a_2^*$ being symmetric,

$$\Delta E = N \int \varepsilon_0(\rho - \rho_{(0)}) = - \sum_i [(1+a_2^*)^{-1} \varepsilon_0](\lambda_{h_i}) = \sum_i \varepsilon(\lambda_{h_i}), \quad (1+a_2^*)\varepsilon = \pi J a_1,$$

which is (19.7) up to πJ : so $\varepsilon = \pi J/(2 \cosh \pi \lambda)$. For the momentum, (15.19) gives $K = \pi M - \frac{2\pi}{N} \sum_j I_j$ exactly, and deleting r roots from a symmetric filled set leaves $\sum_j I_j = -\sum_i I_{h_i}$, so $\Delta K = -\frac{r}{2}\pi + \frac{2\pi}{N} \sum_i I_{h_i}$, and $\frac{2\pi}{N} I_h = \zeta(\lambda_h) = \zeta(-\infty) + 2\pi \int_{-\infty}^{\lambda_h} \rho_t \rightarrow -\frac{\pi}{2} + p(\lambda_h)$, since $\zeta(-\infty) = -(N+r)\pi/2N \rightarrow -\pi/2$ at fixed r . Summing gives $\Delta K = -r\pi + \sum_i p(\lambda_{h_i})$, and r is even. Finally $p' = 2\pi\rho_{(0)} = \pi/\cosh \pi \lambda > 0$ integrates to (19.16), and $\sin p = 1/\cosh \pi \lambda$. ■

The proof is a *backflow* computation. With the bare magnon energy $\varepsilon_0 = -2J/(4\lambda^2 + 1) = -\pi J a_1$, putting r delta-function holes into (19.4) and inverting $1+a_2^*$ dresses the energy of the *removed* magnon, $-\varepsilon_0 = \pi J a_1$ — and that equation is (19.7) again, so $\varepsilon = (-\varepsilon_0)^{\text{dr}} = \pi J \rho$. It runs on *one order finer* information than Theorem 19.3's assumption supplies: at fixed r the hole density tends to zero, and (19.15) needs $\lim N(\rho_N - \rho_{(0),N})$. The careful route subtracts the two finite systems and linearises *before* letting $N \rightarrow \infty$ · cited: Takahashi §3.4; it gives the same ε , and finite chains agree (the check script, $N = 64$ to 1024).

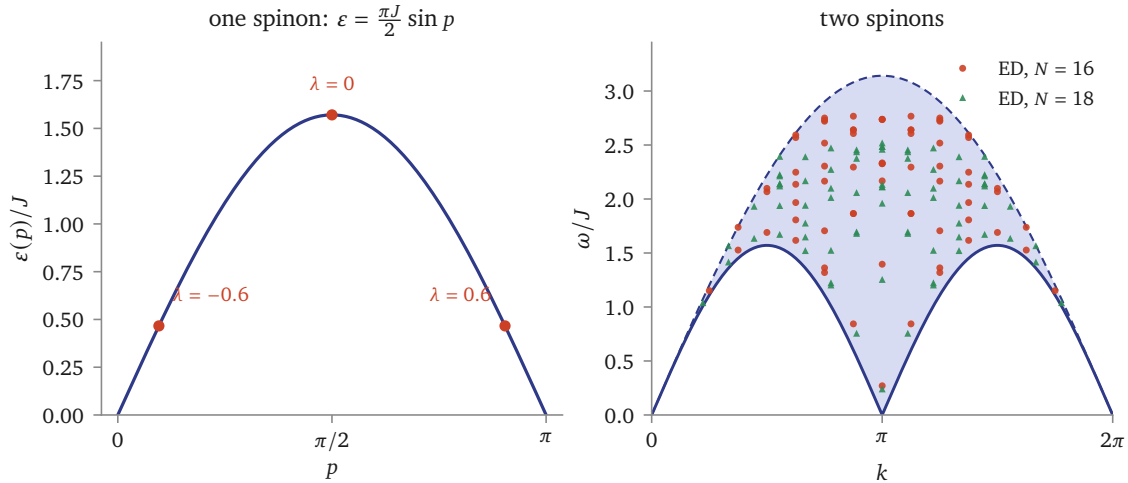


Figure 19.6: Left: the spinon dispersion $\varepsilon = \frac{\pi J}{2} \sin p$ of des Cloizeaux and Pearson, with three rapidities marked; $\lambda = 0$ sits at the top of the band and $\lambda \rightarrow \pm\infty$ at its two ends. Right: the two-spinon continuum, with the exact levels of the $S^z = 1$ sector at $N = 16$ and $N = 18$ that fall inside it. The window is an energy filter, not a classification: at these sizes a few genuine two-hole levels lie just above the dashed thermodynamic boundary and are filtered out along with the four-spinon states.

The continuum, and what a neutron sees

A two-spinon state has $\omega = \varepsilon(p_1) + \varepsilon(p_2)$, $k = p_1 + p_2$, with $p_1, p_2 \in (0, \pi)$ distinct (the holes sit at distinct vacancies). At fixed k , $\sin p_1 + \sin p_2 = 2 \sin \frac{k}{2} \cos \frac{p_1 - p_2}{2}$ grows as $p_1 - p_2$ shrinks and falls as one spinon reaches an end of its band.

Corollary 19.12 (The two-spinon continuum). *The two-spinon states are dense in*

$$\frac{\pi J}{2} |\sin k| \leq \omega \leq \pi J |*| \sin \frac{k}{2}, \quad 0 < k < 2\pi, \quad (19.17)$$

and fill its interior. Neither boundary is attained: the upper one is the limit $p_1 \rightarrow p_2$, the lower one the limit in which one spinon runs to the end of its band, $\lambda \rightarrow \pm\infty$.

The lower boundary is the des Cloizeaux–Pearson dispersion of 1962, read for thirty years as *the* spin-wave branch; the upper one shows it is only the edge of a continuum. That is the measurable difference between a spin chain and a magnet: a three-dimensional antiferromagnet scatters neutrons into a sharp magnon line, the spin- $\frac{1}{2}$ chain into a broad continuum, because a neutron flips one spin and therefore makes spinons in *pairs* — two at least, any even number in principle — which share the energy as they like. Inelastic scattering on KCuF_3 sees such a continuum, and the 1995 measurement was fitted with an approximate two-spinon cross-section, the Müller ansatz. The exact two-spinon contribution, computed later, exhausts about 73% of the integrated intensity; four spinons and more carry the rest · *cited: Tennant, Cowley, Nagler and Tsvetlik, Phys. Rev. B* **52** (1995) 13368; Müller, Thomas, Beck and Bonner, *ibid.* **24** (1981) 1429; Lake et al., *Phys. Rev. Lett.* **111** (2013) 137205.

Lieb’s Type I and Type II

The gas has a Fermi sea with *edges*, so it has two ways to be excited; Lieb found both in 1963, and both come off one pair of functions. The $T = 0$ dressed energy and momentum are

$$\varepsilon(k) - \frac{1}{2\pi} \int_{-q}^q K(k - k') \varepsilon(k') dk' = k^2 - \mu, \quad p'(k) = 2\pi\rho_t(k), \quad (19.18)$$

the first being (19.13) for the bare energy $k^2 - \mu$ and the second Proposition 19.9; the constant μ — the *chemical potential*, and in this lecture there is no spectral parameter to confuse it with — is fixed by $\varepsilon(\pm q) = 0$. That condition does two things at once: it puts both branches at zero energy where they begin, and it is exactly the hypothesis of Proposition 19.9, so the velocity formula applies. Now add a particle just above the right edge, $k > q$: energy $\varepsilon(k) > 0$ and momentum $p(k) - p(q)$ trace the *Type I* branch, the Bogoliubov mode (a particle below $-q$ gives the mirror branch, referred to $-q$). Move a particle from inside the sea to its edge, leaving a hole at $|k| < q$: energy $-\varepsilon(k) > 0$ and momentum $p(q) - p(k)$ trace *Type II*, which rises, turns over and returns to zero at momentum $2\pi n$ · *sketched: the branches are computed here from (19.18); that they exhaust the elementary excitations is cited (Lieb 1963).*

Type II has no counterpart in Bogoliubov theory — it is the exact version of a dark soliton · *cited: Kulish–Manakov–Faddeev (1976)* — and both branches are linear at small momentum with the same slope, the sound velocity that makes the gas a Luttinger liquid.

19.6 The dictionary

Lecture 10 split the real λ -axis of the periodic Toda chain into N bands, where $|t(\lambda)| < 2$ and the two Floquet multipliers sit on the unit circle, separated by *gaps* (Corollary 10.11). The quantum side has just produced the same picture: the rapidities the ground state occupies form a sea, $[-q, q]$ or all of $\mathbb{R}$. In both, a one-dimensional set of spectral parameters is divided into a filled part and an empty part by one function of λ , and in both the *edges* carry

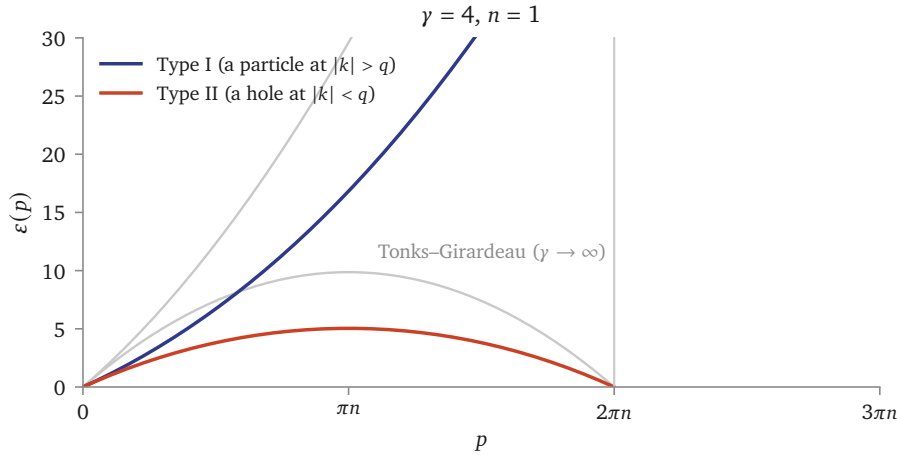


Figure 19.7: Lieb's branches at $\gamma = 4$, $n = 1$. Both leave the origin with the same slope — the dressed velocity v^{eff} (19.14) at the Fermi edge, i.e. the speed of sound. The Type II branch vanishes again at $p = 2\pi n$: the sea has been rotated by one unit. Faint: the Tonks–Girardeau limits $(p + \pi n)^2 - (\pi n)^2$ and $(\pi n)^2 - (\pi n - p)^2$ that the two branches approach.

the physics — a band edge is where the two multipliers collide at $t = \pm 2$, and $\pm q$ is where an excitation costs nothing.

The second row is sharper. [Corollary 5.4](#) found an invariant torus's frequencies as $\nu = \partial K / \partial I$: the rate at which a classical motion runs along it. The quantum analogue is the dressed velocity v^{eff} (19.14). Neither is a bare $d\epsilon/dk$; both are derivatives *with everything else held fixed*, which is what makes them the velocities transport uses — as [Lecture 23](#) needs.

Dictionary

The bands and gaps of the periodic Toda chain's spectral parameter ([Corollary 10.11](#)): the real λ -axis split by $|t(\lambda)| < 2$, with the physics at the edges $\rightarrow$ the Fermi sea of rapidities and its edges $\pm q$ (19.10), where the dressed energy vanishes (type A)

The frequency $\nu = \partial K / \partial I$ of a Liouville torus ([Corollary 5.4](#)) $\rightarrow$ the dressed velocity $v^{\text{eff}} = d\epsilon/dp$ (19.14) (type A; both sides proved, the link is the correspondence principle)

Reading

19.7 The two ends of $e(\gamma)$, and the central charge

The asymptotics of § 19.3 continue

$$e(\gamma) = \gamma - \frac{4}{3\pi} \gamma^{3/2} + \left(\frac{1}{6} - \frac{1}{\pi^2}\right) \gamma^2 + O(\gamma^{5/2}), \quad e(\gamma) = \frac{\pi^2}{3} \left(1 - \frac{4}{\gamma} + \frac{12}{\gamma^2} + O(\gamma^{-3})\right), \quad (19.19)$$

· *cited for the coefficients* and the check script confirms each where [Figure 19.5](#) draws it, the strong-coupling residual matching the known third coefficient in size and sign. It also pins the curve at an independently published point: by Hellmann–Feynman the kinetic and interaction energies are equal at $\gamma = 4.3485(1)$, where $e/2 = 0.82668(1)$ — and the solver used here gives 4.3484 and 0.82665.

Figure 19.3 was a straight line in $1/N^2$, and its slope is not an accident. A one-dimensional critical system in a box of N sites has

$$E_0(N) = Ne_0 - \frac{\pi c \nu}{6N} + o(N^{-1}), \quad (19.20)$$

with ν the sound velocity and c the central charge of the conformal field theory that describes its long distances · *cited: Blöte–Cardy–Nightingale and Affleck (1986)*. For the Heisenberg antiferromagnet (19.16) gives $d\varepsilon/dp = (\pi J/2) \cos p$, so the speed of sound — its value as $p \rightarrow 0$ — is $\nu = \pi J/2$, and (19.20) predicts $E_0(N)/N = -J \ln 2 - \pi^2 J/(12N^2)$ if $c = 1$. That is the slope in Figure 19.3. Reading it backwards, the numbers $-(E_0 - Ne_0) 6N/\pi$ fall from 1.596 at $N = 12$ to 1.583 at $N = 20$ by exact diagonalisation and reach 1.5721 at $N = 400$ by the Bethe equations, against $\pi J/2 = 1.5708$: *the central charge of the spin- $\frac{1}{2}$ chain is 1*, measured from its ground-state energies alone. The Lieb–Liniger gas gives $c = 1$ the same way · *cited*.

This is the cheapest known route to a central charge, and it is why the finite-size Bethe equations — solvable to machine precision at $N = 400$ — are a standard instrument rather than a curiosity.

Tutorial

19.8 Solving Lieb’s equation

Theorem 19.6 is a Fredholm equation of the second kind, and the Nyström method turns it into linear algebra. Choose an n -point Gauss–Legendre rule on $[-1, 1]$, scale it to $[-q, q]$ as $k_i = qx_i$, $w_i \mapsto qw_i$, and impose (19.10) at the nodes:

$$\left(\delta_{ij} - \frac{w_j}{2\pi} K(k_i - k_j) \right) \rho_j = \frac{1}{2\pi}, \quad n \approx \sum_i w_i \rho_i, \quad \frac{E}{L} \approx \sum_i w_i k_i^2 \rho_i. \quad (19.21)$$

The kernel is analytic in the separation, so for a fixed $\alpha = c/q$ the rule converges faster than any power of n : at $\gamma \approx 1$, $n = 20$ already gives machine precision. But the constant in front of that statement is α , and the rule resolves a spike of width α only when n is large compared with $1/\alpha$. Measured against an $n = 1600$ reference:

q/c	γ	$n = 10$	$n = 40$	$n = 200$
0.1	29	$2 \cdot 10^{-13}$	$2 \cdot 10^{-13}$	$2 \cdot 10^{-13}$
1.3	1.2	$6 \cdot 10^{-7}$	$3 \cdot 10^{-13}$	$3 \cdot 10^{-13}$
20	$9 \cdot 10^{-3}$	$7 \cdot 10^2$	0.7	$6 \cdot 10^{-8}$
200	10^{-4}	$6 \cdot 10^6$	$2 \cdot 10^5$	$5 \cdot 10^3$

At the last row the computed density is not merely inaccurate: it goes *negative*, which the true one cannot (the Neumann series has a positive source and a positive kernel). *Always run n against $2n$* . A rule that is exact to machine precision in one regime can be useless two decades away, and nothing in the output says so.

One more subtlety: the nodes never reach $\pm q$, so any quantity wanted *at the edge* — the chemical potential of (19.18), or the sound velocity — must be evaluated by putting $k = q$ back into the integral equation rather than by extrapolating the nodes. Interpolating with the *linear* rule that comes to hand put μ wrong in the fourth decimal, where the integral equation gives it beyond the sixth; a high-order interpolant through all the nodes loses

nothing. The lesson is not that interpolation is bad but that the edge value is free — the equation already evaluates there.

The pair that shares a quantum number. Remark 19.2's counterexample is worth meeting. At $N = 22$, $M = 2$,

$$\lambda_1 = 2.228461914574\dots, \quad \lambda_2 = 2.372786474870\dots, \quad I_1 = I_2 = \frac{19}{2}, \quad (19.22)$$

solve (19.1) to forty digits, and Lecture 15's two-magnon wavefunction built on them is a genuine eigenvector: $\|\Psi\|^2 = 5.7312$ and $\|H\Psi - E\Psi\|/\|\Psi\| = 10^{-14}$ at $E = -0.1808904556J$. Between the two roots $\zeta' = -4.6 \times 10^{-4}$. Such pairs first appear at $N \approx 21.9$, which is why nothing at the $N \leq 20$ of Figure 19.3 could have found one: *when a claim quantifies over all N , test the smallest N at which its failure mode is known to appear, not the largest N the machine allows.*

Which way does p run? Theorem 19.11 could as well have printed $p = 2\pi \int_{\lambda}^{\infty} \rho = \frac{\pi}{2} - \arctan \sinh \pi \lambda$: it satisfies $\varepsilon = \frac{\pi J}{2} \sin p$ just as well, and for a pair of holes at $\pm \lambda$ gives the same total, π . A symmetric test cannot tell the two apart — which is exactly why one must test an asymmetric pair. Against finite chains with the holes placed asymmetrically the printed orientation is right to 10^{-4} at $N = 256$ and the other is wrong by 1.13. The counterexample lives in the check script.

To get $e(\gamma)$ one does not solve for q at fixed γ : by the scaling in the proof of Theorem 19.6 it is easier to set $c = 1$, sweep q , and read off the pair $(\gamma, e) = (1/n, (E/L)/n^3)$. The curve of Figure 19.5 is 260 such solves and takes about a second.

The same solve answers a question that looks harder. By Hellmann–Feynman the interaction energy is $\langle 2c \sum \delta \rangle = c \partial E / \partial c$ at fixed N, L , so per particle it is $n^2 \gamma e'(\gamma)$ and the kinetic energy is $n^2(e - \gamma e')$. They are equal when $e = 2\gamma e'$, i.e. when the slope of $\ln e$ against $\ln \gamma$ is $\frac{1}{2}$; the sweep above puts that at $\gamma = 4.3484$ with $e/2 = 0.82665$, against the published $\gamma = 4.3485(1)$ and $0.82668(1)$ · cited: Caux, The Bethe Ansatz (*integrability.org*), the Lieb–Liniger ground state. A benchmark with two independent digits is worth more than a plausible-looking curve: this one is what fixed which power of n the normalisation (19.11) carries.

Summary.

- *Vacancies:* $N\zeta/2\pi$ sweeps an interval of length $N - M$, so an M -magnon state has $N - M$ quantum numbers, and $N - 2M$ of them are holes *when the roots are distinct and ζ increases* (Proposition 19.1; Remark 19.2 shows the hypothesis is needed) — on an even ring, holes in pairs.
- *The density equation* (19.4) and its Lieb–Liniger twin (19.10) are one equation (19.5) with kernels of opposite sign (19.6) (Remark 19.4); the limit producing them is an assumption (Theorem 19.3).
- *The antiferromagnet:* $\rho = 1/(2 \cosh \pi \lambda)$ by Fourier and $e_0 = -J \ln 2$ (Theorem 19.5).
- *The gas:* Lieb's equation (19.10) on $[-q, q]$, one function $e(\gamma)$ (19.11), solved numerically; at $\gamma \rightarrow \infty$ it is a gas of free continuum fermions (Proposition 19.7), closing Lecture 14's loop; at $\gamma \rightarrow 0$, Bogoliubov's $\gamma - \frac{4}{3\pi} \gamma^{3/2}$.
- *Dressing* (19.13): one linear operator for every quantity, and a *response*, not a total. $2\pi \rho_t = (p'_0)^{\text{dr}}$; at $T = 0$, and only where the dressed energy vanishes at the sea's edges, $v^{\text{eff}} = d\varepsilon/dp$ (19.14) (Proposition 19.9).

- *Spinons*: one missing root, two holes, spin $\frac{1}{2}$ each; $\varepsilon = \frac{\pi J}{2} \sin p$ (Theorem 19.11) and the continuum (19.17) that neutron scattering measures. For the gas, Lieb’s Type I and Type II branches.

Tutorial. Exercise Sheet 19.

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